

Anas Al Dahamsheh

AI Engineer | LLM, RAG, AI Agents & Full-Stack Development

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PROFESSIONAL SUMMARY

AI Engineer who builds production-ready large language model (LLM) applications end to end: retrieval-augmented generation (RAG), AI agents with tool calling, chatbots, automation and LLM evaluation, backed by full-stack delivery in Python (FastAPI) and TypeScript (React, Next.js). At ITC International, builds AI projects, chatbots and automation, red-teams and evaluates AI models, and creates training data. Independently designed and shipped 10 bilingual (English/Arabic) applications, including a hybrid-search research assistant that ranks the correct passage first 93.9% of the time with a 54 ms median search over ~97,000 passages, and an agentic platform that generates verified fine-tuning datasets. Test-driven, with a trust-and-safety background that shapes reliable, responsible AI.

PROFESSIONAL EXPERIENCE

AI Engineer | ITC International

Dec 2025 – Present

- Build LLM-powered applications, chatbots and Python automation workflows end to end, from prompt and pipeline design to integration and testing.
- Create training and evaluation datasets for AI models, writing challenging multilingual prompts across formats, languages and Arabic dialects to target known and new failure modes.
- Design adversarial test scenarios (AI red teaming) that probe LLM reliability, factual accuracy, hallucination, bias and unsafe behavior.
- Evaluate model outputs against a structured rubric (accuracy, relevance, completeness, neutrality, safety, source validation) and trace the root cause of each failure with supporting evidence.
- Deliver reproducible defect reports (prompt, response, weakness, recommended correction) that drive model improvements and technical investigation.

Content Security Moderator | BIGO Live

Apr 2023 – Dec 2025

- Reviewed live-stream and user-generated content, identifying harmful material and policy violations and applying community guidelines to keep the platform safe.
- Worked with the team to sharpen policy interpretation and reporting practices, experience that now underpins AI safety and evaluation work.

SELECTED PROJECTS

Noesis | Bilingual RAG Research Assistant with Verified Citations

Python, FastAPI, React, TypeScript, SQLite FTS5, USearch (HNSW), Gemini API, embeddings, GitHub Actions

- Engineered hybrid bilingual retrieval (BM25 + dense embeddings in an HNSW index + offline vectors) fused with reciprocal rank fusion (RRF) and MMR: correct passage ranked first 93.9% of the time, other-language document 90%, 54 ms median search over ~97,000 passages.
- Built a tool-calling research agent (6 tools) that streams cited answers on ~3K input tokens per answer, plus a deterministic sentence-level citation verifier for English and Arabic that needs no extra model call.
- Added a knowledge graph, a grounded Q&A dataset miner with JSONL/Alpaca export, JWT auth with rotating refresh tokens, role-based access and an audit trail; 133 automated tests (97 backend, 36 frontend) in CI.

Corpusforge | Agentic Training-Data Platform for LLM Fine-Tuning

Python, FastAPI, async SQLAlchemy, Next.js, TypeScript, Gemini API, OpenAI-compatible APIs, SSE

- Designed a planning agent (function calling) that turns a plain-language goal into a costed job, then generates multi-turn conversations grounded in uploaded files, imported URLs or web research.
- Built a quality gate of deterministic checks (grounding, language, privacy, variety), feedback-driven repair loops, deduplication and 0–100 scoring: a real job delivered 8/8 accepted conversations (average quality 84.7) in 3 min 11 s for \$0.02.
- Implemented adaptive rate limiting (sliding request/token windows, AIMD concurrency, 429 back-off), resumable jobs, live SSE monitoring and export to OpenAI JSONL, ShareGPT, Alpaca and Parquet; runs on Gemini or any OpenAI-compatible endpoint (OpenAI, Ollama, vLLM).

Jubran & Abu Jbara | AI Waiter and QR Ordering Platform for Two Amman Restaurants

Python, FastAPI, Next.js, TypeScript, PostgreSQL, pgvector, WebSockets, Docker, Gemini and OpenAI APIs

- Built a bilingual AI waiter chatbot with tool calling and semantic menu search (pgvector, 266 embedded documents for a 229-dish menu) that places orders only after server-side validation and explicit guest confirmation.
- Delivered a real-time staff floor and guest order tracking over WebSockets with a transactional outbox, Alembic migrations and Docker Compose; manual ordering keeps working during AI-provider outages; 262 automated tests.

Sieve | Adaptive Web Scraping and Data Extraction Engine

Python, FastAPI, httpx, Playwright, trafilatura, SQLite FTS5, Next.js, WebSockets, Gemini API

- Designed an 8-step site probe that selects HTTP/2, a browser-grade TLS client or headless Chromium per site and escalates automatically; one job crawled 290 pages across 4 site types in 2 min 13 s.
- Built an NLP cleaning pipeline (boilerplate learning, Arabic normalization, language detection, SHA-256 + SimHash near-duplicate removal, PII redaction), LLM-extracted typed JSON fields and an AIMD resource governor that keeps CPU under its ceiling.

Vigil | Website Health Checker at Scale

Python, FastAPI, dnspython, httpx, Playwright, Next.js, TypeScript, SSE, GitHub Actions

- Built layered DNS, TCP, TLS and HTTP verification with real-browser escalation, returning 9 verdicts with confidence scores and evidence; an 85-link audit finished in 52 s with every verdict correct.
- Raised throughput from ~1 to 6–7 URLs per second with adaptive concurrency; 226 automated tests (182 backend, 44 frontend) on Ubuntu and Windows CI.

CV Checker | ATS Resume Analyzer with Grounded LLM Generation

Next.js, TypeScript, Gemini API (structured output), Zod, Vitest, Playwright, axe-core

- Built an explainable, rule-based ATS scoring engine and requirement-to-evidence job matching; LLM output is schema-validated with grounding checks and a deterministic fallback, and no resume data is ever stored.

More projects: Exam Maker (AI interview simulator with AES-256-GCM sealed answer keys and rubric-based LLM grading), Giant Store (e-commerce with a RAG shopping assistant), Website QA Agent (Playwright crawler for SEO, accessibility and security audits; Turborepo, BullMQ, Redis). Live demos: anas-ai-portfolio.netlify.app

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS

Generative AI & LLMs: LLM application development, RAG, AI agents, agentic workflows, tool/function calling, chatbots, prompt engineering, structured outputs, LangChain, LangGraph, LlamaIndex, Model Context Protocol (MCP), Gemini API, OpenAI API, Ollama

Machine Learning: PyTorch, Hugging Face Transformers, scikit-learn, fine-tuning (LoRA), NumPy, pandas

LLM Evaluation & Safety: AI red teaming, LLM evaluation, hallucination and grounding checks, citation verification, rubric-based scoring, bias and safety testing, training data creation

Retrieval, NLP & Data: Embeddings, vector search, hybrid search (BM25 + dense), HNSW, pgvector, RRF, MMR, Arabic NLP, language detection, near-duplicate detection, web scraping, fine-tuning dataset generation

Backend: FastAPI, Node.js, REST APIs, async Python, Pydantic, SQLAlchemy, Alembic, WebSockets, server-sent events, JWT authentication, rate limiting, automation

Frontend: React, Next.js, Tailwind CSS, Zustand, TanStack Query, responsive English/Arabic (RTL) interfaces

Databases & DevOps: PostgreSQL, SQLite, Redis, Prisma, Docker, Docker Compose, AWS (EC2, S3, Lambda), Git, GitHub Actions (CI/CD), Netlify, Vercel

Testing: pytest, Vitest, Playwright, end-to-end testing, accessibility testing (axe-core)

EDUCATION

Bachelor of Science in Computer Engineering | Al-Balqa Applied University

2025

Faculty of Engineering Technology, Amman, Jordan

CERTIFICATIONS AND TRAINING

- Full Stack Development Training (130 hours) | Digital Opportunity Trust (DOT) Jordan | May 2026 | Credential ID: DOT-AM-FS-2-25-YGA-An-0526
- Retrieval-Augmented Generation for Enhanced AI Outputs | IBM SkillsBuild | Sep 2026
- Generative AI Essentials: Using LLMs to Work with Data | IBM SkillsBuild | Sep 2026
- Craft Precise Prompts for AI Models | IBM SkillsBuild | Sep 2026
- Getting Started with Generative AI | IBM SkillsBuild | Sep 2026

LANGUAGES

Arabic (Native) | English (Professional working proficiency)